Prompting for AI-Assisted Coding and Analysis

Nathan J. McNeese

Associate Vice President for Technology & Innovation
McQueen Quattlebaum Endowed Professor
Founding Director, Clemson Center for Human-AI Interaction, Collaboration, & Teaming
Founding Director, TRACE Research Group

The Importance of Prompts

Prompting coordinates intent and action

In human–AI work, prompting translates a scientific objective into a bounded contribution—connecting the shared goal to work the AI can attempt and the human can inspect.

- State the analytical goal and the decision it supports.
- Supply the context the AI cannot safely infer.
- Ask for one immediate, bounded task.
- Define constraints and what must remain unchanged.
- Specify the evidence or checks required before acceptance.

Prompts are structured instructions

A prompt is a structured work request. It reduces ambiguity by connecting the desired outcome, available evidence, permitted actions, and definition of a useful response.

- Goal: What outcome or decision are we trying to reach?
- Context: What does the AI need to know about the data and setting?
- Task: What should it do now?
- Constraints: What boundaries must it respect?
- Deliverable: What should come back?
- Validation: How will the result be checked?

Effective prompting is analytical clarity—not a collection of magic words.

Good prompts produce inspectable work

The strongest prompts make the AI's contribution visible enough to review. Length matters less than whether the requested work has clear boundaries and produces evidence.

- Produce one bounded result rather than an entire pipeline.
- Separate assumptions, observations, and interpretations.
- Show proposed changes before applying consequential edits.
- Surface unresolved questions instead of inventing answers.
- Return code, checks, or artifacts the analyst can independently inspect.

How to Articulate Prompts

Think before asking

Prompting starts after the analyst has framed the scientific problem. AI can help clarify choices, but it should not silently decide what outcome matters or how the forecast will be used.

Define before prompting:
- Research or public-health question.
- Population, location, and unit of analysis.
- Outcome variable and reporting definition.
- Forecast horizon and intended use.
- Known limitations, constraints, and decision stakes.

Define the scientific objective

Specific objectives produce more relevant code and more defensible outputs.

Weak

“Find interesting patterns and forecast the disease.”

Stronger

“Forecast weekly reported cases for each location one through four weeks ahead to support short-term resource planning.”

What improved?

Target • unit • horizon • intended use

Role clarity: some decisions remain human

Effective teaming requires clear decision authority. AI can support these decisions, but it should not silently make them.

Human responsibility

- Outcome definition
- Scientific interpretation
- Privacy and ethics
- Final claims

AI can support

- Alternative options
- Code implementation
- Diagnostics
- Documentation

Do not delegate accountability with the task.

Choose the tool before the prompt

Different AI surfaces have different context, capabilities, and risk.

Conversation

- Concepts and explanations
- Small code examples
- Critique and planning

Editor assistant

- Current files and selections
- Focused code changes
- Immediate feedback

Coding agent

- Multi-file work
- Terminal and tests
- Stronger boundaries required

More autonomy changes the request

A coding agent needs operational boundaries that a conversational answer may not require.

For a response

- Desired explanation
- Code format
- Assumptions
- Evidence

For an agent

- Files it may change
- Commands it may run
- Tests required
- Approval and stopping conditions

AI context comes from several places

AI tools do not share one universal context window. What a system can use depends on the interface, attached files, open workspace, permissions, and current conversation.

- Current prompt and prior conversation.
- Open files, selected code, or active editor context.
- Repository files and project instructions.
- Terminal output, test results, and error messages.
- Attached data dictionaries or documentation.
- Tool-provided reference material and available commands.

Shared understanding requires visible context

Human and AI cannot coordinate from different assumptions. Before asking for analysis or code, confirm that the AI has the correct and current information.

- Is the correct dataset, file, or code selection available?
- Are the target, unit, and forecast horizon defined?
- Is the data dictionary current?
- Are permitted packages and file boundaries explicit?
- Does the AI know which decisions are settled and which remain open?

Missing context becomes hidden uncertainty.

Clear instructions reduce ambiguity

AI fills missing details with patterns from training—not necessarily with your intended assumptions.

Tell the AI

- The problem
- The relevant inputs
- What to do now
- What not to do

Also define

- Desired output
- Quality criteria
- Required checks
- Unresolved questions

Writing the Actual Prompt

A strong prompt answers eight questions

The components can be brief, but they should be deliberate.

Purpose

1. Goal
2. AI role
3. Context

Work

4. Inputs
5. Immediate task
6. Constraints

Evidence

7. Deliverable
8. Validation

Prompt structure should match task complexity—not a fixed formula.

State the desired outcome

Describe the decision or state you want to reach—not a vague activity.

Weak

“Help with this dataset.”

Better

“Determine whether this surveillance dataset is structurally ready for weekly time-series modeling.”

A good goal

Names the decision • narrows the scope • supports evaluation

Give the AI a bounded role

A bounded role tells the AI which function to perform in the current step. It focuses the response without pretending the model has credentials, authority, or accountability it does not possess.

Examples:

- Data-quality reviewer: identify issues and propose checks.
- R coding assistant: implement an approved change.
- Statistical critic: challenge assumptions and evaluation design.
- Reproducibility reviewer: inspect dependencies and execution steps.
- Documentation assistant: explain accepted code and decisions.

Provide operational definitions

Column names rarely communicate the reporting process, scientific meaning, or exceptions that determine how analytical code should behave.

Include:
- What one row represents.
- Date, population, and geographic units.
- Outcome definition and measurement process.
- Missing-value and zero-count conventions.
- Reporting delays, revisions, and backfills.
- Known anomalies or structural changes.

Clear next actions coordinate the work

A bounded next task is a coordination move: it defines one observable AI contribution and makes the human’s next inspection point clear.

Useful task verbs:

- Inspect or summarize the current state.
- Compare specified alternatives.
- Generate a bounded code change.
- Diagnose a known error or unexpected result.
- Test an assumption with explicit checks.
- Critique or document an existing artifact.

Constraints prevent silent divergence

Constraints establish the analytical and operational boundaries of the work.

Analytical

- Preserve chronological order
- Do not impute without approval
- Compare with a baseline
- Do not change the target

Operational

- Use R only
- Use approved packages
- Modify one named file
- Ask before restructuring

Specify what should come back

An output contract makes the result easier to inspect and compare.

Analysis	Code	Communication
• Prioritized issue table • Comparison table • Diagnostic summary	• Runnable R script • Proposed patch • Tests or assertions	• Assumptions • Questions • Change summary

Build checking into the request

Verification should be part of the request, not an afterthought. A useful AI response reveals the assumptions it made and provides evidence a human can independently examine.

Request:

- Assumptions and unresolved questions.
- Tests, assertions, or diagnostic checks.
- Expected outputs or success conditions.
- Potential failure points and edge cases.
- Comparison with a simple baseline.
- A concise change summary tied to the request.

A Prompt Pipeline

The mega-prompt hides decisions

One request for the entire pipeline gives the AI too much implicit authority.

Too broad

“Clean the data, choose the best model, forecast the next month, and write the report.”

Better start

“Audit the dataset first. Do not modify data or build a model. Return issues, questions, and R checks.”

Why stage it?

Reviewable outputs • fewer hidden assumptions • easier recovery

Prompt the pipeline through checkpoints.

Inspect before cleaning

Begin with diagnosis. Ask the AI to identify possible data problems and generate checks without giving it permission to rewrite the dataset or start modeling.

Ask it to examine missingness, duplicates, gaps in time, inconsistent locations, impossible values, and reporting artifacts. Require:

- A prioritized issue table.
- Questions that require human judgment.
- R code that verifies each suspected issue.
- No data modification and no model fitting.

Make cleaning decisions explicit

After the analyst approves specific decisions, the AI can implement them in code. The prompt should preserve raw data and make every transformation visible.

Require the code to:

- Implement only approved cleaning steps.
- Record row counts before and after each step.
- Report affected records and unexpected values.
- Flag unexplained observations instead of deleting them.
- Produce a concise transformation log.

Describe patterns before explaining them

Use visualization to describe structure before asking the AI to explain it. Patterns in surveillance data may reflect disease dynamics, reporting processes, or both.

Ask for weekly case-count plots by location that:

- Preserve chronological order.
- Show missing reporting periods.
- Mark flagged or revised observations.
- Use consistent axes when comparison matters.
- Separate descriptive observations from causal interpretations.

Ask the AI to critique the work

Generation proposes a result; critique searches for reasons that result may fail. Asking for a separate critique changes the interaction from production to adversarial review.

- Check for future leakage and invalid time splits.
- Identify unjustified statistical or epidemiological assumptions.
- Flag implausible predictions and unstable behavior.
- Find conclusions that exceed the evidence.
- Prioritize issues by likely impact and propose specific checks.

A critique is evidence for the next human review—not proof that the work is wrong.

TAKEAWAY

Before you send a prompt

Clear prompts create bounded outputs that humans can run, inspect, and improve.

Goal → Context → Task → Check

A prompt is a coordination move: ask for the next piece of work—not an unreviewed final analysis.